Synoptic CB: Manual Redox Probes

2023 Samples

2026-04-14

Measurements taken with SWAP Redox Probes & Calomel Ref probe

Protocol: https://docs.google.com/document/d/

1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit

Units = mV

##Add Required Packages

##Setup

```
###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
dat <- read.csv("Raw Data/COMPASS_Synoptic_CB_Redox_2023_Raw.csv")
soildat <- read.csv("Raw Data/CB_Soil_pH.csv")

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_Synoptic_CB_Redox_2023_processed.csv"

##Fix Sample Zone and Site naming
soildat <- soildat %>%
  mutate(Zone = case_when(Site == "SWH" & Zone == "Upland" ~ "Swamp",
                          Site == "SWH" & Zone == "Upland-Control" ~ "Upland",
                          TRUE ~ Zone),
         Site = ifelse(Site == "Gcrew", "GCW", Site))

##Fix Site naming
dat <- dat %>%
  mutate(Zone = case_when(Site == "SWH" & Zone == "Upland" ~ "Swamp",
                          Site == "SWH" & Zone == "Upland Control" ~ "Upland",
                          Site == "SWH" & Zone == "Upland-Control" ~ "Upland",
                          TRUE ~ Zone),
         Site = ifelse(Site == "Gcrew", "GCW", Site),
         Month = ifelse(Month == "Arpil", "April", Month))

dat$Site <- gsub(" ", "", dat$Site) ## Remove space in GWI site name
```

```r
##Add column for SWH topography
dat <- dat %>%
  mutate(Topography = case_when(Site == "SWH" & Zone == "Transition-Hummock" ~ "Hummock",
                                Site == "SWH" & Zone == "Transition-Hollow" ~ "Hollow",
                                TRUE ~ NA))

##Check redox depths of 30 cm
dat30 <- dat %>%
  filter(Depth == 30)

##Std error function
std.error <- function (x, na.rm=FALSE){
  sqrt(var(x, na.rm = na.rm) / sum(!is.na(x)))}
```

Adjust Raw Readings by pH

Need to adjust for pH:

redox = redoxi + [(ph - 5)59 + 224] This is the equation from Megonigal,

Patrick, & Faulkner 1993

They used a calomel electrode and used 224 to adjust to H2,

we use a KCl ref probe, so we need to add 202 instead

```r
##Remove unneccesary column in redox data
prelong <- dat %>%
  select (-Campaign) %>%
  mutate(Notes = NA,
         Time_24hr = NA,
         Time_Zone = NA)

#switch redox data to long format
datlong <-  as.data.frame(melt(setDT(prelong), id.vars = c("Date", "Month", "Year", "Site", "Zone", "Dep
                                variable.name = "RP_mV"))
datlong$Site <- gsub(" ", "", datlong$Site) ## Remove space in GWI site name

# ##Add in soil pH data
# all_corr <- merge(datlong, soildat, by = c("Site", "Zone"), all = TRUE)
#
# ##Adjust redox for pH
# all_corr <- all_corr %>%
#   mutate(value_corr = value + ((pH - 5)*59 + 202))

##Average across all 5-6 measurements for each sampling Site, Zone, Depth, Month, Year
```

```r
red1 <- datlong %>%
  group_by(Date, Month, Year, Site, Zone, Depth, Time_24hr, Time_Zone, Topography, Notes) %>%
  summarise(Date = mean(Date),
            Avg_mV = mean(value, na.rm=TRUE),
            Std_dv = sd(value, na.rm=TRUE),
            Std_err = std.error(value, na.rm=TRUE),
            Notes = first(Notes)
            )


# head(red1)
```

##Write Out Adjusted & Avg'd Data

Vizualize Data